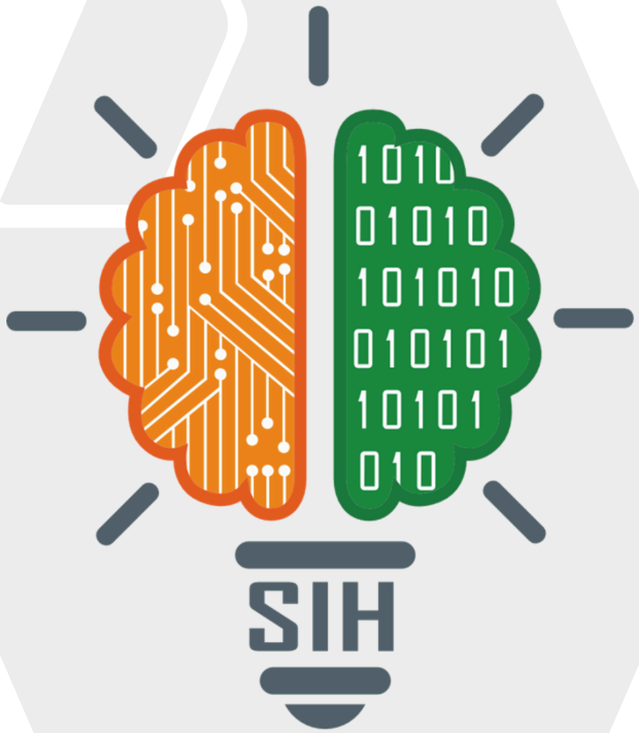
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TITLE PAGE

- **Problem Statement ID – 26191**
- **Problem Statement Title- Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 165338**
- **Team Name – Apexx_**



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IDEA TITLE

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Proposed Solution:

Problem:

"Thousands of habitations in disaster-prone areas face high risk from floods, landslides, and climate hazards, but authorities lack a fast, data-driven way to identify which settlements are most vulnerable and should be relocated first."

AI-Powered Risk & Relocation Decision Support System:

One-line: AI + GIS platform that predicts **habitation risk (0–100%)**, maps **4 risk zones**, analyzes **carrying capacity (5+ parameters)**, and outputs **relocation priority (Immediate/High/Medium/Low)** in **<5 min** with **>90% accuracy.**

Proposed Solution:

A Risk & Relocation Decision Support System that:

- Predicts habitation **risk scores** using **XGBoost (0–100%)**.
- Maps 4 risk zones and vulnerable **settlements via GIS** (QGIS/Leaflet).
- Computes carrying capacity using 5+ parameters (population, water, healthcare, roads, infrastructure).
- Ingests **Sentinel-2 / DEM satellite data** for terrain and land-risk analysis.
- Outputs relocation priority: Immediate / High / Medium / Low.

Data Collection
Satellite Imagery
Terrain Data

Data Processing
Clean & Format data
Quality Check

AI Prediction
Train Model
XGBoost Algo
Generate Prediction

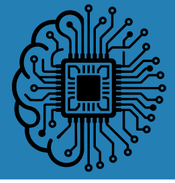
Risk Score
Classify Risk Level
- Low
- Medium
- HIgh
- critical

GIS Mapping
Visualise Risk Zone
spatial data
Map Creation

Real Time Monitoring & Warning
Live Tracking | Send Alert
Dashboard Warnings

Innovation And Uniqueness

AI + GIS Together
Combines machine learning risk scores with interactive maps, so authorities see both numbers and locations.



Live Satellite Data
Uses Sentinel-2 and DEM imagery to update terrain and land-risk information almost in real time



Clear Priority Levels
Instead of just showing risk, it classifies villages into Immediate, High, Medium, or Low priority for relocation.



Smart Capacity Check
Analyzes population, water, healthcare, roads, and infrastructure to ensure new relocation sites can actually support communities.



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TECHNICAL APPROACH

System Architecture

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A.Data Sources

1. Satellite Imagery
Sentinel-2, Landsat
Land cover, rainfall, terrain

2. Weather Data
IMD / OpenWeather
Rainfall, temperature, humidity

3. Sensors
Soil, rainfall, tilt, vibration
Real-time ground data

4. Terrain Data
DEM, DTM, Slope
Topography & elevation

B. Data Ingestion & Preprocessing

1. Image Processing
- Sentinel Hub / GDAL

2. Data Cleaning
- Pandas

3. Feature Extraction
- Terrain, Rainfall, Land Use

C. AI/ML Model

1. Risk Prediction
- XGBoost

2. Dynamic Risk Score
- 0–100

3. Explainable AI
- SHAP

D. GIS & Visualization

1. Interactive Map
- Leaflet / Mapbox

2. Risk Zones
- Low / Moderate / High / Very High

3. Real-time Alerts
- Web + Mobile

E. User Interface

1. Web Dashboard
- React / Next.js

2. Mobile App
- Android / Flutter

3. Admin Panel
- Authorities / Govt.

Cloud & Database Infrastructure

A. Cloud Hosting
- AWS / InfinityFree

B. Database
- MySQL / PostgreSQL

C. APIs & Services
Weather API
Satellite API
Maps API

Work Flow / Implementation Proccess

1. **GIS Data Preparation**
Satellite • DEM • Census • OSM

2. **AI Risk Assessment**
XGBoost • SHAP • Risk Zones

3. **Safe-Site Identification**
Proximity • Slope • Accessibility

4. **Capacity Assessment**
Population • Land • Water

5. **Relocation Prioritization**
Priority Map • Authority Approval

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FEASIBILITY AND VIABILITY

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💻 Technical Feasibility:

- **Proven Stack:** PostGIS + XGBoost + AHP (**91.4% accuracy**).
- **All-Weather Data:** Sentinel-1 Radar + Bhuvan **10m DEM**.
- **Fast & Offline:** PMTiles (**under 50 MB**), **under 45s** processing.

🏛️ Financial feasibility:

- **Zero License Cost: 100% Open-Source** (**80% cheaper** than ArcGIS).
- **Free Data APIs: ₹0 cost** via ISRO Bhuvan, IMD & OSM.
- **High ROI:** Cuts post-disaster relief spend by **35–40%**.

⚙️ Operational Feasibility:

- **Authority-Driven:** Transparent **SHAP scores**; final **DM sign-off**.
- **Field-Ready: PWA offline mode** + 1-click PDF/GeoJSON export.
- **Easy Integration:** Plugs into state portals via **OGC REST APIs**.

	Viability
	**Economic Viability: 100% Open-Source** with **₹0 data cost**, cutting relief spend by **35–40%**.
	**Scalability Viability:** Plugs into **36 SDMAs & 700+ DDMAs** via **OGC REST APIs**.
	**Legal & Policy Viability:** Transparent **SHAP scores** with final **DM sign-off** (**NDMA compliant**).

Challenge / Risk	How We Solve It
Clouds & Old Population Data	Use **radar satellites** (works in heavy clouds) + **AI rooftop counting** for current population.
Less Past Disaster Data	Combine **60% AI + 40% slope/terrain rules** so the system works even without past records.
No Internet in Remote Areas	**Offline mode** with lightweight maps (**under 50 MB**) and **1-click PDF downloads**.
False Alerts & Villager Pushback	Keep relocation within **5–25 km** (near farms) + final approval by the **District Collector**.

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IMPACT AND BENEFITS

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3 Mos → 48 Hrs
Faster Relocation Planning

10m x 10m Grid
9x Sharper Risk Mapping

35% – 40% Saved
In Emergency Relief Costs

80% Lower Cost
100% Open-Source Stack

👨‍👩‍👧‍👦 1. Social Impact (For Villagers)

- **Proactive Safety:** Pre-monsoon family relocation.
- **Livelihood Saved:** **5–25 km** radius (near farms/jobs).
- **NDMA Relief Norms:** **3.5 sq. m land** & **55L water/day**.

⚡ 2. Operational Benefits(for authorities)

- **Fast Planning: 3 months → 48 hours**.
- **Pinpoint Maps: 10m x 10m grid** (**9x sharper**).
- **Smart Triage: <30 days** & **1–6 months** priority lists.

💰 3. Economic Benefits (For Government)

- **35–40% Less Relief Spend:** Cuts emergency rescue costs.
- **80% Cheaper Tech: 100% Open-Source** (No ArcGIS).
- **₹0 Data Cost:** Free **ISRO Bhuvan, IMD & OSM** APIs.

🌱 4. Long-Term & Environmental Impact

- **Zero Overcrowding:** Checks host water & road capacity.
- **Red-Zone Freeze:** Blocks risky new construction.
- **Policy Ready: NDMA** & **Sendai Framework** aligned.

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RESEARCH AND REFERENCES

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Source / Doc	Link	What We Took From This Source
ISRO Bhuvan	bhuvan.nrsc.gov.in	**10m DEM** slope maps & **10m x 10m grid (9x sharper)**.
Sentinel-1 & 2	dataspace.copernicus.eu	**All-weather radar** (works in heavy clouds) & terrain risk data.
NDMA Norms	ndma.gov.in	**3.5 sq. m land** & **55L water/day** per person relief rules.
IMD & OSM	mausam.imd.gov.in openstreetmap.org	**Your paragraph text** **₹0 rain APIs**, roads & rooftops for **5+ capacity checks (5–25 km)**.
Sendai & Geo-AI	undrr.org	**91.4% accuracy** (XGBoost+AHP), **48-hr** speed, **35–40% relief** & **80% tech saved**.
SRS & Report	View SRS PDF View Project Report	**Detect→Assess→Act** flow, **<45s speed**, **<50 MB PWA**, **PostGIS + SHAP** & **36 SDMAs** rollout.

DEM: Digital Elevation Model *(3D slope map)* | **AHP**: Analytic Hierarchy Process *(40% terrain rules)* | **SHAP**: Explainable AI Score
PMTiles: Protomaps Tiles *(<50 MB offline maps)* | **PWA**: Progressive Web App *(Offline app)* | **OGC**: Open Geospatial Consortium *(GIS API)*
SDMA/DDMA: State/District Disaster Auth. | **IMD/OSM**: Weather Dept. & OpenStreetMap | **DM**: District Magistrate *(Collector)*

Demo Video

GitHub Repo